

WTW158 Memo - Worksheet on integration 2

1. $\int x^4 dx = \frac{1}{5}x^5 + c$
2. $\int (7 - 6x)^4 dx = -\frac{1}{30}(7 - 6x)^5 + c$
3. $\int \sin^4(5x) \cos 5x dx = \frac{1}{25} \sin^5 5x + c$
4. $\int \sqrt{x} dx = \frac{2}{3}x^{\frac{3}{2}} + c$
5. $\int x^3 \sqrt{x^4 + 5} dx = \frac{1}{6}(x^4 + 5)^{\frac{3}{2}} + c$
6. $\int \frac{\sqrt{\ln x}}{x} dx = \frac{2}{3}(\ln x)^{3/2} + c$
7. $\int e^{-5x} dx = -\frac{1}{5}e^{-5x} + c$
8. $\int e^{\cos 3x} \sin 3x dx = -\frac{1}{3}e^{\cos 3x} + c$
9. $\int e^{3x} \sin(e^{3x}) dx = -\frac{1}{3} \cos(e^{3x}) + c$
10. $\int_{-2}^{-1} \frac{1}{x^4} dx = \frac{7}{24}$
11. $\int 3^x dx = \frac{1}{\ln 3} 3^x + k$
12. $\int \frac{4^x}{(1 - 4^x)^6} dx = \frac{1}{5(\ln 4)(1 - 4^x)^5} + k$
13. $\int \sec^2(2^x) 2^x dx = \frac{1}{\ln 2} \tan 2^x + c$
14. $\int \operatorname{cosec}(3x) \cot(3x) dx = -\frac{1}{3} \operatorname{cosec}(3x) + c$
15. $\int x \operatorname{cosec}^2 x^2 dx = -\frac{1}{2} \cot x^2 + c$
16. $\int \frac{\sec(\ln x) \tan(\ln x)}{x} dx = \sec(\ln x) + c$
17. $\int 3 dt = 3t + c$
18. $\int \frac{1}{t^5} dt = -\frac{1}{4t^4} + c$
19. $\int \frac{1}{\sqrt{z}} dz = 2\sqrt{z} + c$
20. $\int_1^3 \sin(\pi x) dx = 0$
21. $\int \frac{2}{1 + 4x} dx = \frac{1}{2} \ln|1 + 4x| + c$
22. $\int \frac{2x}{1 + 4x^2} dx = \frac{1}{4} \ln|1 + 4x^2| + c$
23. $\int \frac{2}{1 + 4x^2} dx = \arctan 2x + c$

24. $\int \frac{x^4}{1+x^{10}} dx = \frac{1}{5} \arctan x^5 + c$
25. $\int \frac{1}{\sqrt{1-9x}} dx = -\frac{2}{9} \sqrt{(1-9x)} + c$
26. $\int \frac{1}{\sqrt{1-9x^2}} dx = \frac{1}{3} \arcsin 3x + c$
27. $\int \frac{x^3}{\sqrt{1-9x^4}} dx = -\frac{1}{18} \sqrt{1-9x^4} + c$
28. $\int \frac{x}{\sqrt{1-9x^4}} dx = \frac{1}{6} \arcsin 3x^2 + c$
29. $\int \frac{2+x}{\sqrt{1-x^2}} dx = \int \frac{2}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx =$
 $2 \arcsin x - \sqrt{1-x^2} + k$
30. $\int_{-2}^{-1} \frac{1}{x} dx = -\ln 2$
31. $\int \frac{t^9 \ln(1+t^{10})}{1+t^{10}} dt = \frac{1}{20} \ln^2 |1+t^{10}| + c$
32. $\int \frac{\cos 2u}{\sin 2u} du = \frac{1}{2} \ln |\sin 2u| + c$
33. $\int \frac{\sin(\ln v)}{v} dv = -\cos(\ln v) + c$
34. $\int \sec^2(e^{2t}) e^{2t} dt = \frac{1}{2} \tan e^{2t} + c$
35. $\int 3^{\tan 4z} \sec^2 4z dz = \frac{1}{4 \ln 3} 3^{\tan 4z} + c$
36. $\int \left(\frac{3}{x} - \frac{x}{3} \right) dx = 3 \ln |x| - \frac{1}{6} x^2 + c$
37. $\int \frac{2+x^4}{x^2} dx = \int \left(\frac{2}{x^2} + x^2 \right) dx = -\frac{2}{x} + \frac{1}{3} x^3 + c$
38. $\int \frac{x}{2+x^4} dx = \frac{\sqrt{2}}{4} \arctan \frac{x^2}{\sqrt{2}} + c$
39. $\int x^2 \sqrt{x} dx = \int x^{\frac{5}{2}} dx = \frac{2}{7} x^{7/2} + c$
40. $\int_{-2}^{-1} |x| dx = \int_{-2}^{-1} -x dx = \frac{3}{2}$

ERROR: syntaxerror
OFFENDING COMMAND: --nostringval--

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